Multilevel Domain Decomposition-Based Architectures for Physics-Informed Neural Networks

Aeden Lindelauf, Daniel Iordanov

Eindhoven University of Technology

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Abstract

Physics-Informed Neural Networks (PINNs) offer an elegant way to approximate solutions to differential equations. In particular, they are trained by governing the underlying physical laws directly into the training objective. In practice, however, they might experience slow convergence or poor approximations on large and complex domains. To counteract this phenomenon, multilevel Finite Basis Physics-Informed Neural Networks are architected, such that they decompose the domain into overlapping subdomains. In this paper, we reproduce several experiments from Dolean et al. (2024) on the 2D Laplacian problem. More specifically, we investigate the effect of number of levels, overlap ratio, and network width. Our results confirm that multilevel Finite Basis Physics-Informed Neural Networks outperform standard PINNs in terms of stable approximation quality and computational efficiency. Additionally, results show that increasing the width of the subdomain networks typically enhances approximation quality. Finally, we perform additional experiments on the 1D Poisson equation involving high-frequency components, which shows that multilevel FBPINNs are capable of learning these components if sufficiently small subdomains are used.

1 Introduction

Differential equations, would they be ordinary, partial, or stochastic, are the cornerstone of most mathematical models in physics, engineering, biology, and quantitative finance. Yet, most of them cannot be solved analytically. There are classical numerical methods, such as finite differences or finite elements, which aim to approximate their solutions numerically. In essence, they act as a discretiser of the domain on a mesh, which transforms the continuous problem into a large linear system. Those methods are well-understood and optimized; however, they often become extremely expensive for high-dimensional domains or complex geometries. Physics-Informed Neural Networks (PINNs) are an alternative to these classical numerical schemes, e.g., one can consult Raissi et al. (2019). In this framework, a neural network is being trained while satisfying the underlying physical laws. In particular, the differential equation and the boundary conditions are directly included in the training objective. The structure of PINNs allows for a natural extension to tackle, for example, inverse problems or work efficiently in high-dimensional settings. As with every model, however, PINNs have a certain set of limitations (Moseley et al. (2023)). For example, training a single neural network to approximate a solution over a large domain can lead to slow convergence and reduced accuracy. This is especially true when the solution exhibits high-frequency oscillations. Finite Basis Physics-Informed Neural Networks (FBPINNs) aim to tackle this challenge. The main idea is that the domain is split into overlapping subdomains, whereby each one is handled by a local, smaller network. This framework is proposed such that it makes each local problem easier and, in that way, tries to tackle the inability of standard PINNs to capture high-frequency oscillations, also known as Spectral bias. These FBPINNs can be combined with multilevel domain decomposition methods, as proposed by Dolean et al. (2024). The goal of this research is to investigate the proposed method and to determine the reproducibility of the results in Dolean et al. (2024).

2 Methodology

The basic idea of the methodology proposed by Dolean et al. (2024) is the idea of using PINNs in combination with domain decomposition (DD) techniques to improve both accuracy and convergence while solving differential equations. This section will cover the theory of PINNs, FBPINNs, and explain the idea of multilevel FBPINNs.

2.1 Physics-Informed Neural Networks

First, we discuss the theory behind PINNs. Consider the following boundary value problem (BVP),

$$\begin{aligned} \mathcal{N}[u](x) &= f(x), \quad x \in \Omega \subset \mathbb{R}^d \\ \mathcal{B}_k[u](x) &= g_k(x), \quad x \in \Gamma_k \subset \partial\Omega, \end{aligned}$$

where $\mathcal{N}[\cdot]$ is some (non)linear differential operator, $\mathcal{B}_k[\cdot]$ is a set of boundary conditions (BCs) applied to u on some segment of the boundary Γ_k, and $u(\cdot)$ is the solution to this BVP (Dolean et al. (2024)). Physics-informed neural networks (PINNs) approach this problem by approximating the solution u with a neural network, i.e., $u(x, \theta) \approx u(x)$, where θ are the learnable parameters. The loss function, which is used to train the PINN, is then given in terms of the BVP as follows,

$$\begin{aligned} \mathcal{L}(\theta) = &\frac{\lambda_I}{N_I} \sum_{i=1}^{N_I} \underbrace{\left(\mathcal{N}[u](x_i, \theta) - f(x_i)\right)^2}_{\text{PDE residual}} \\ &+ \sum_{k=1}^{N_k} \frac{\lambda_B^k}{N_B^k} \underbrace{\left(\mathcal{B}_k[u](x_i^k, \theta) - g_k(x_i^k)\right)^2}_{\text{BC residual}}. \end{aligned}$$

Here, $\{x_i\}_{i=1}^{N_I}$ is a set of collocation points samples in the interior of the domain, $\{x_j^k\}_{j=1}^{N_B^k}$ is a set of points sampled on the boundary Γ_k and λ_I and λ_B^k are scalars that put more weight on the PDE residual or the residual on the BCs. Standard techniques from machine learning are then used to minimise this loss function, thus finding a numerical approximation to the BVP. Intuitively, this works because minimising the PDE residual is tantamount to ensuring that the neural network satisfies the underlying PDE, and similarly, by minimising the BC residual, the network is forced to obey the BCs as well.

2.2 Finite Basis PINNs

Finite Basis Physics-Informed Neural Networks adapt the so-called divide-and-conquer approach. In particular, the global solution is represented by a smaller neural network defined over overlapping subdomains. For the purpose, let $\Omega \subset \mathbb{R}^d$ denote the global domain, $\Omega_j \subset \Omega$ denote the overlapping sub-domains for all $j \in \{1, \ldots, J\}$, where $J \in \mathbb{N}$ denotes the number of overlapping sub-domains. Then, we have that

$$\Omega = \bigcup_{j=1}^{J} \Omega_j.$$

Importantly, note that in our experiments we have that neighbouring sub-domains always overlap, i.e., $\Omega_i \cap \Omega_j \neq \emptyset$. Next, we define the so-called window functions ω_j such that $\text{supp}(w_j) \subset \Omega_j$. As in Dolean et al. (2024), they are chosen such that they form a partition of unity, such that

$$\sum_{j=1}^{J} w_j(x) \equiv 1, \qquad \forall x \in \Omega.$$

Therefore, we can now define the global FBPINN approximation. It is given by

$$u(x, \theta) = \sum_{j=1}^{J} \omega_j(x) \nu_j(x, \theta_j).$$

To summarise, we train PINNs on subdomains on Ω_j and then 'average' the result using the window function.

2.3 Multilevel FBPINNs

In the previous section, FBPINNs are defined on *a* set of subdomains. However, instead of only using one domain decomposition, multilevel FBPINNs consider several of them simultaneously. First, we define L levels of DDs, as in previous section, where each level ℓ defines an overlapping DD of Ω with J^ℓ subdomains. In other words, the DD of level ℓ can be written as $D^\ell = \{\Omega_j^\ell\}_{j=1}^{J^\ell}$ for $1 \leq \ell \leq L$. As for FBPINNs, we define a neural network on each subdomain ν_j^ℓ for $j = 1, \ldots, J^\ell$ and $1 \leq \ell \leq L$. Furthermore, we also define a window function $\omega_j^{(\ell)}$ for each subdomain $\Omega_j^{(\ell)}$ in each level ℓ. The global space approximation is then given by

$$u(x, \theta) = \frac{1}{L} \sum_{\ell=1}^{L} \sum_{j=1}^{J^{(\ell)}} \omega_j^{(\ell)} \nu_j^{(\ell)}(x, \theta_j^{(\ell)}).$$

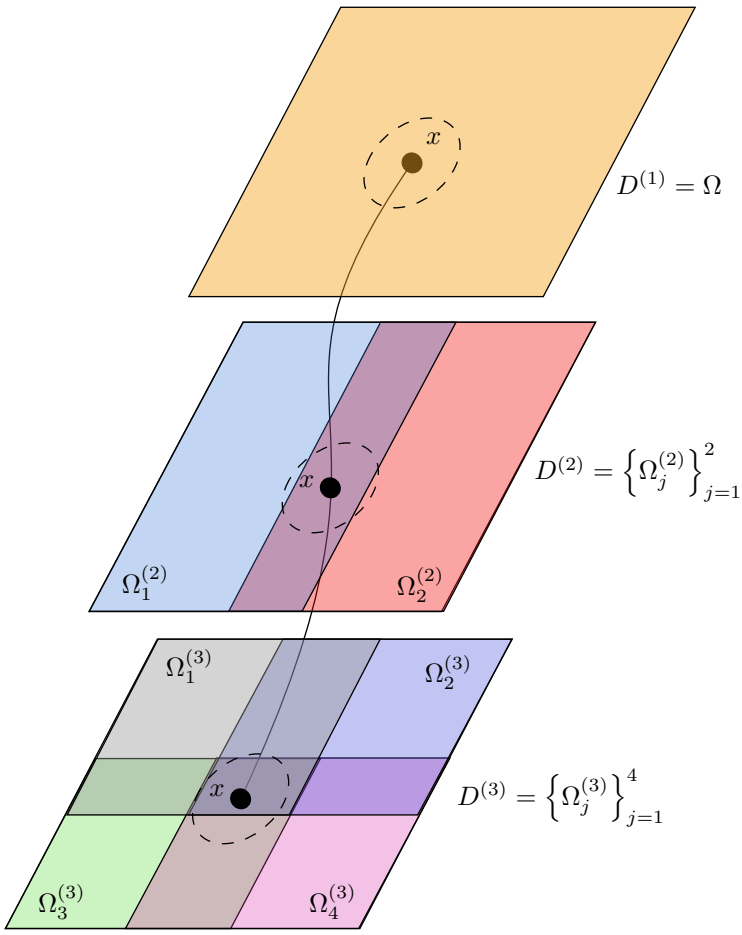


Figure 1: Visualisation of a multilevel FBPINN. On each subdomain $\Omega_j^{(\ell)}$, we define a PINN. Then, using window functions, we can construct an FBPINN on each domain $D^{(\ell)}$. Intuitively, to compute the value in the point x is tantamount to taking the 'average' over the dashed oval in each level. The value in the point x, computed by the multilevel FBPINN, is then given by the average of the three FBPINNs value of x.

This approach is intuitively visualised in Figure 1. In this figure, three levels have been defined with each level having its own DD $D^{(\ell)}$, $1 \leq \ell \leq 3$. For each level, we train an FBPINN $u^{(1)}$, $u^{(2)}$ and $u^{(3)}$ on each DD $D^{(1)}$, $D^{(1)}$ and $D^{(3)}$. To compute the value in a point x, we then compute the average of the FBPINNs

$$u(x, \theta) = \frac{u^{(1)}(x, \theta^{(1)}) + u^{(2)}(x, \theta^{(2)}) + u^{(3)}(x, \theta^{(3)})}{3}.$$

The idea behind this approach is that the multilevel FBPINN learns the higher frequency components as well, since these are converted into lower frequencies by the DD and normalisation of the domain. This in turn should then reduce the spectral bias often visible in PINNs.

3 Reproduced results

In this section, we reproduce the ablation experiments for the 2D homogeneous Laplace problem, which are mentioned in the paper from Dolean et al. (2024).[1] In our experiment, we test the performance of multilevel FBPINNs by adjusting the architecture choices such as the number of levels, the overlap ratio, denoted by δ, between neighbouring subdomains, as well as the size of the subdomain networks. In the paper of Dolean et al. (2024), this is used as a benchmark to assess the qualitative and quantitative behaviour of the architecture. Subsequently, they move to a more challenging Helmholtz example. Here, we only consider the 2D homogeneous Laplace problem that is defined as

$$\begin{cases} -\Delta u = f & \text{in } \Omega = [0,1]^2, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

where f is given by

$$f(x_1, x_2) = 32(x_1(1-x_1) + x_2(1-x_2)).$$

The exact solution, for this particular choice of f, is

$$u(x_1, x_2) = 16(x_1(1-x_1)x_2(1-x_2)).$$

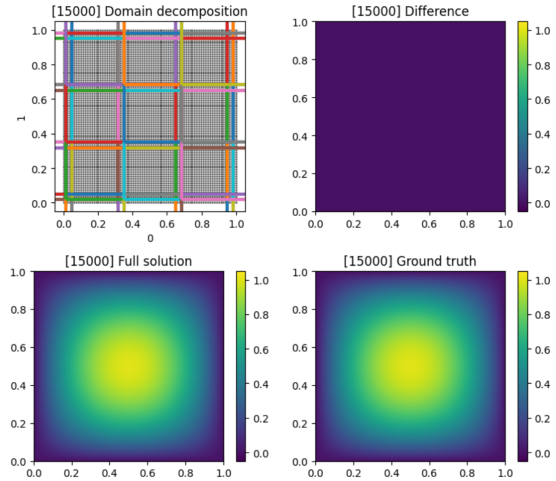
The baseline configuration of our model is a multilevel FBPINN with $L = 3$ levels, overlap ratio $\delta = 1.9$, fully connected subdomain networks with 1 hidden layer and 16 hidden units. The model is trained using a hard constraining operator

$$[\mathcal{C}u](x, \theta) = \prod_{i=1}^{2} \tanh\left(\frac{x_i}{\sigma}\right) \tanh\left(\frac{1-x_i}{\sigma}\right) u(x, \theta),$$

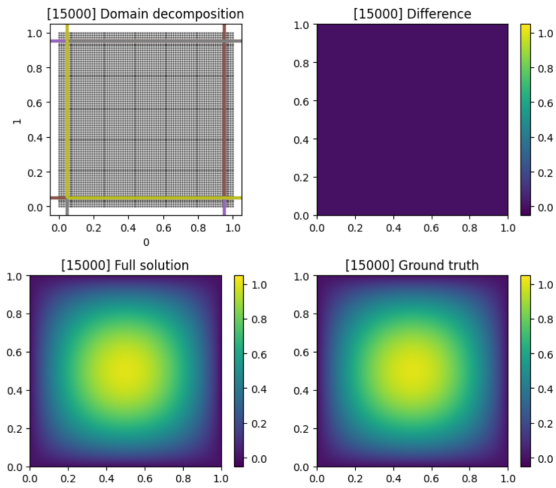
where $\sigma = 0.2$. Furthermore, the number of collocation points sampled from the domain equal $80 \times 80 = 6\,400$, and the model was evaluated on a 350×350 test grid. Finally, we train each model using 15 000 training steps in comparison to the 20 000 steps used as per Dolean et al. (2024), due to the high computational costs it incurs for our experiments. First, we consider the effect of the number of levels. In particular, we test four different decompositions, namely, $[2], [4], [1, 2]$, and $[1, 2, 4]$. The list here represents the number of subdomains within each level. Figure 2 presents the predicted solutions of the homogeneous 2D Laplace problem for multilevel FBPINN with levels [1,2,4] and levels [1,2]. The remainder are available in the Jupyter notebooks published on GitHub.

We now proceed by investigating how convergence is affected by the four different decompositions. In particular, we consider the normalised L^1 test loss. Figure 3 illustrates precisely this. We note that all models successfully approximate the solution to the PDE. However, in general, the multilevel architecture achieves better results throughout the training process. This is in parity with what is observed in the paper by Dolean et al. (2024). Namely, that multilevel structures help with information exchange within the different subdomains.

[1] The code can be accessed on GitHub.



(a) Multilevel FBPINN with levels [1,2,4].



(b) Multilevel FBPINN with levels [1,2].

Figure 2: Predicted solutions of the homogeneous Laplacian problem using multilevel FBPINN architectures.

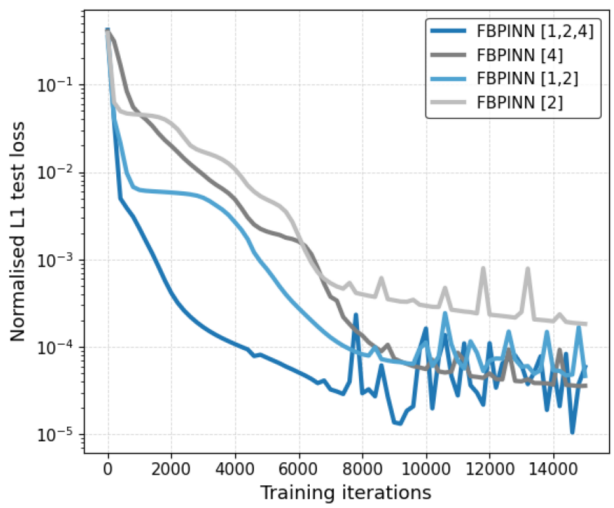


Figure 3: Normalised L^1 test error during training for different FBPINN domain decompositions: $[1, 2, 4]$, $[4]$, $[1, 2]$, and $[2]$.

Next, we continue by investigating how convergence is affected by the overlap ratio δ, while using the baseline multilevel configuration $[1, 2, 4]$. The values we consider for δ are $1.1, 1.5, 1.9,$ and 2.3. The convergence curves are available in Figure 4. Again, all models perform sufficiently well in terms of the L^1 normalized test loss. However, to be more specific, we note that increasing the overlap generally improves the error during training and improves both stability and accuracy. Nevertheless, at the final step (15 000), the results remain fairly close. Thus, in that sense, it is difficult to infer which δ performs the best.

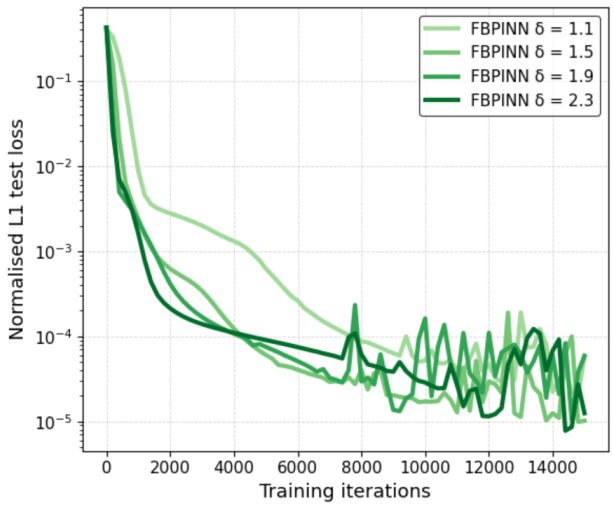


Figure 4: Normalised L^1 test error during training for different multilevel FBPINN overlap ratios δ, $1.1, 1.5, 1.9,$ and 2.3. and fixed multilevel configuration $[1, 2, 4]$.

Finally, we study how the different hidden widths affect performance. In particular, we used the baseline multilevel architecture $[1, 2, 4]$ along with $\delta = 1.9$, and trained FBPINNs consisting of one hidden layer with $2, 4, 8,$ and 16 hidden units, respectively. As per Dolean et al. (2024), we trained a standard PINN with three hidden layers, each consisting of 64 units. Figure 5 shows the convergence curves of these models. It can be noted that, as expected, increasing the width of the subdomain networks improves the quality of our approximations throughout the training process. We can see that up to step 2 000, the PINN baseline outperforms the more involved models; however, from then on, it stagnates and exhibits unstable behaviour until the end of the training process. In comparison, the multilevel FBPINNs have better approximation quality, and as we increase the width of the subdomain networks, it gets further improved. Thus, the benefit of the domain decomposition architecture is evident. To further strengthen this conclusion, we also look at the training efficiency as shown in Figure 6. This statistic measures the number of training steps that are processed per second. This is a good indicator of computational efficiency. The results clearly show that the baseline PINN has a lower training rate compared to the more sophisticated methods. In the performed experiment, training the PINN required more than one hour, while the other models finished training within 15 minutes at most. This is, of course, in parity with the expectations, since it is much harder for a baseline PINN to approximate the solution over the entire domain.

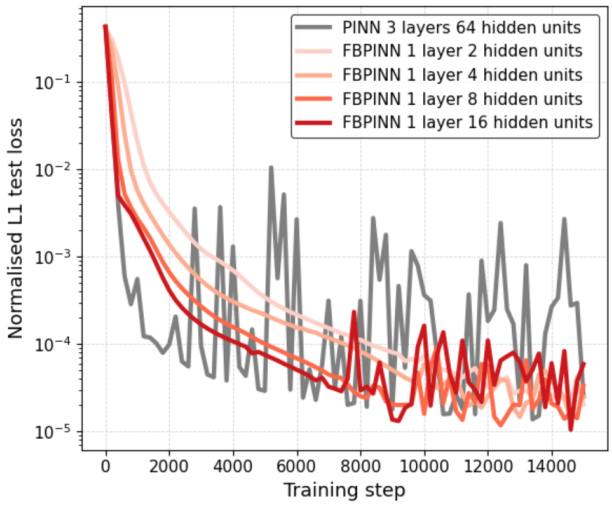


Figure 5: Normalised L^1 test error during training for FBPINNs with different hidden layer widths using the baseline architecture $[1, 2, 4]$ and $\delta = 1.9$, compared to a standard PINN with three hidden layers of 64 units.

Figure 6: Training efficiency for FBPINNs with different hidden layer widths compared to a standard PINN with three layers of 64 units.

4 Critical analysis

In this section, we will evaluate the proposed method critically. First, we will use a (multi-level) FBPINN to solve the 1D Poisson Equation, which is the following ordinary differential equation (ODE):

$$u''(x) = f(x), \quad x \in \Omega = (0, 1), \quad u(0) = u(1) = 0,$$

for some function f to be defined. In this case, we would like to investigate how well (multi-level) FBPINNs solve a differential equation containing high-frequency components.

Therefore, f is defined as follows,

$$f(x) = \sum_{i=1}^{n} a_i \sin(b_i \pi x),$$

with $n \in \mathbb{N}^+$ and $a_i, b_i \in \mathbb{R}^+$ for $i = 1, \ldots, n$. Using the linearity of the differential operator, the exact solution is given by

$$u(x) = -\sum_{i=1}^{n} \frac{a_i}{\pi^2 b_i^2} \sin(b_i \pi x).$$

In this particular example, we choose $n := 2$, $(a_1, a_2) := (12\,500, 1\,000)$ and $(b_1, b_2) := (25, 10)$. The reason for the large amplitudes is due to the integration, which, for reasonable b_i, yields a very small amplitude otherwise. We then construct several multilevel FBPINNs and one PINN with three hidden layers, each consisting of 64 nodes, which we use to solve the BVP. The results are shown in Figure 7. One can immediately observe that the FBPINN with multilevels [2,4,8,16] and [16] performs best. This is, of course, no surprise since these multilevels use smaller DD, thus allowing the (multilevel) FBPINN to learn the high frequency components better. Interestingly, the multilevel $[2, 4, 8, 16]$ performs better than [16]. This is also in agreement with the results from Dolean et al. (2024). Furthermore, we can also conclude that the multilevels with smaller DD do not have the capability to learn the high-frequency oscillations.

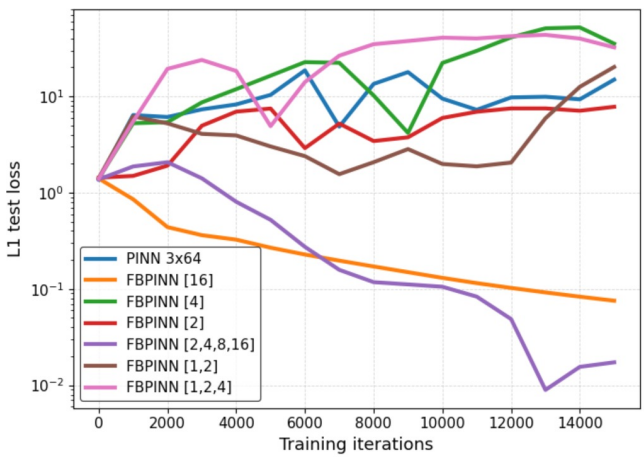


Figure 7: This figure shows the L^1 test loss for general (multilevel) FBPINNs and one PINN with three hidden layers, each consisting of 64 nodes.

To display this effect, three solutions to the BVP are shown in Figure 8. This figure shows that both the PINN and the FBPINN with multilevel [1,2,4] are not able to solve the BVP, whereas this is the case for the FBPINN applied to 16 subdomains.

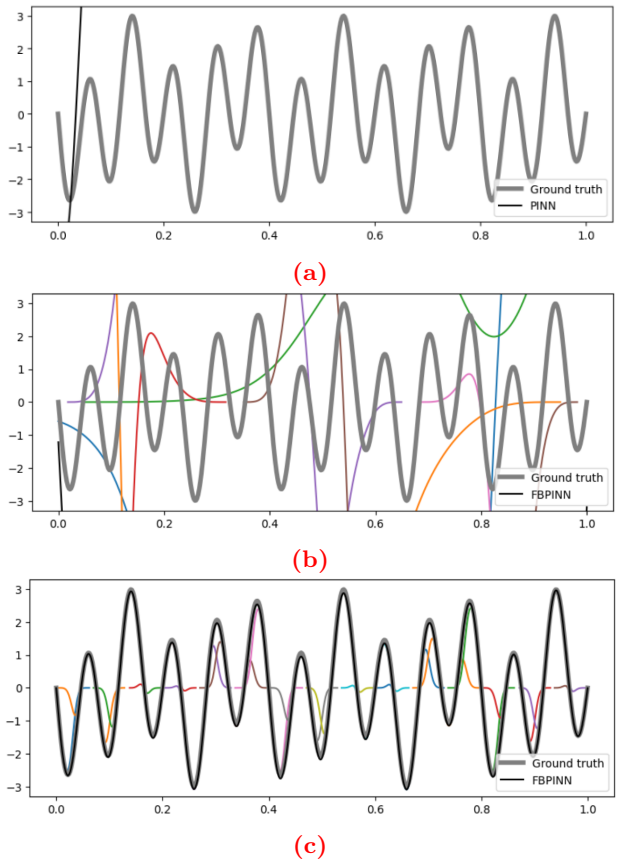


Figure 8: In figure (a), the solution of a PINN, consisting of three hidden layers with 64 nodes in each layer, is displayed. In figure (b), the FBPINN solution, with multilevel [1,2,4] configuration, are displayed together with the solutions on the subdomains (the non-black/gray curves). In figure (c), the FBPINN solution, applied to 16 subdomains, are displayed together with the solutions on the subdomains (the non-black/gray curves).

From the previous results, it becomes apparent that, for multi-scale problems involving high-frequency components, a large domain decomposition is necessary. However, this poses the first challenge. In the context of a BVP, one requires a priori knowledge of the spectral behaviour of the solution to determine an appropriate DD capable of learning higher frequency components. However, if such a DD is used, then we recognise that FBPINNs outperform PINNs if a sufficient DD is chosen. Also, when comparing the multilevel FBPINNs with, for example, the FNO model, another drawback with the current implementation becomes apparent. The goal of the FNO is to be trained on a set of input parameters, such as varying the PDE parameters or initial conditions, and to use the trained model to infer the solution to new situations. This is not the case with multilevel FBPINNs. If the inputs to the BVP are altered, the multilevel FBPINNs need to be retrained on the newly formulated problem. To overcome this challenge, Dolean et al. (2024) propose to incorporate BCs and PDE coefficients and to train across a range of these inputs so that the multilevel FBPINN does not have to be retrained for each new instance (Dolean et al., 2024). The final observation is by Dolean et al. (2024), which corresponds to the performance of

multilevel FBPINNs on BVPs with complex geometries. In this research, a two-dimensional rectangle is used on which the Laplace equation is solved. However, one could also consider a star-shaped domain. Although the current framework could handle such geometries, implementing an optimal DD and window function may be too challenging (Dolean et al., 2024).

5 Conclusion

The purpose of this article is to research the multilevel domain decomposition-based architectures for physics-informed neural networks by Dolean et al. (2024). The first part focused on investigating the reproducibility of the method for the Laplace equation. We conclude that multilevel FBPINNs outperform (FB)PINNS in terms of accuracy and stability, and are faster. This is consistent with the findings by Dolean et al. (2024). The second part involved testing the multilevel FBPINN method in an experiment involving high-frequency components. The results suggest that, provided the domain is decomposed into sufficiently small subdomains, multilevel FBPINNs can learn these high-frequency components. However, several challenges remain, such as the need for a priori knowledge of the spectral behaviour of the solution, adapting to more complex geometries, and including BCs and PDE coefficients in the training of the FBPINN to avoid the need for retraining with each new problem. The latter two are proposed in Dolean et al. (2024).

One idea that might be worth exploring is replacing the neural network within the PINN with a Fourier kernel. Fourier kernels convert input channels to the frequency domain and are therefore potentially suitable for solving multi-scale problems. This approach might address the issue of unknown spectral behaviour of the solution.

In summary, the multilevel FBPINNs are an effective method for solving multi-scale phenomena, but there are several open challenges for future research.

References

Dolean, V., Heinlein, A., Mishra, S., and Moseley, B. (2024). Multilevel domain decomposition-based architectures for physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 429:117116.

Moseley, B., Markham, A., and Nissen-Meyer, T. (2023). Finite basis physics-informed neural networks (fbpinns): a scalable domain decomposition approach for solving differential equations. *Advances in Computational Mathematics*, 49(62).

Raissi, M., Perdikaris, P., and Karniadakis, G. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707.